

HASTH VANI

Prototype Build Plan — Illustrated Directions

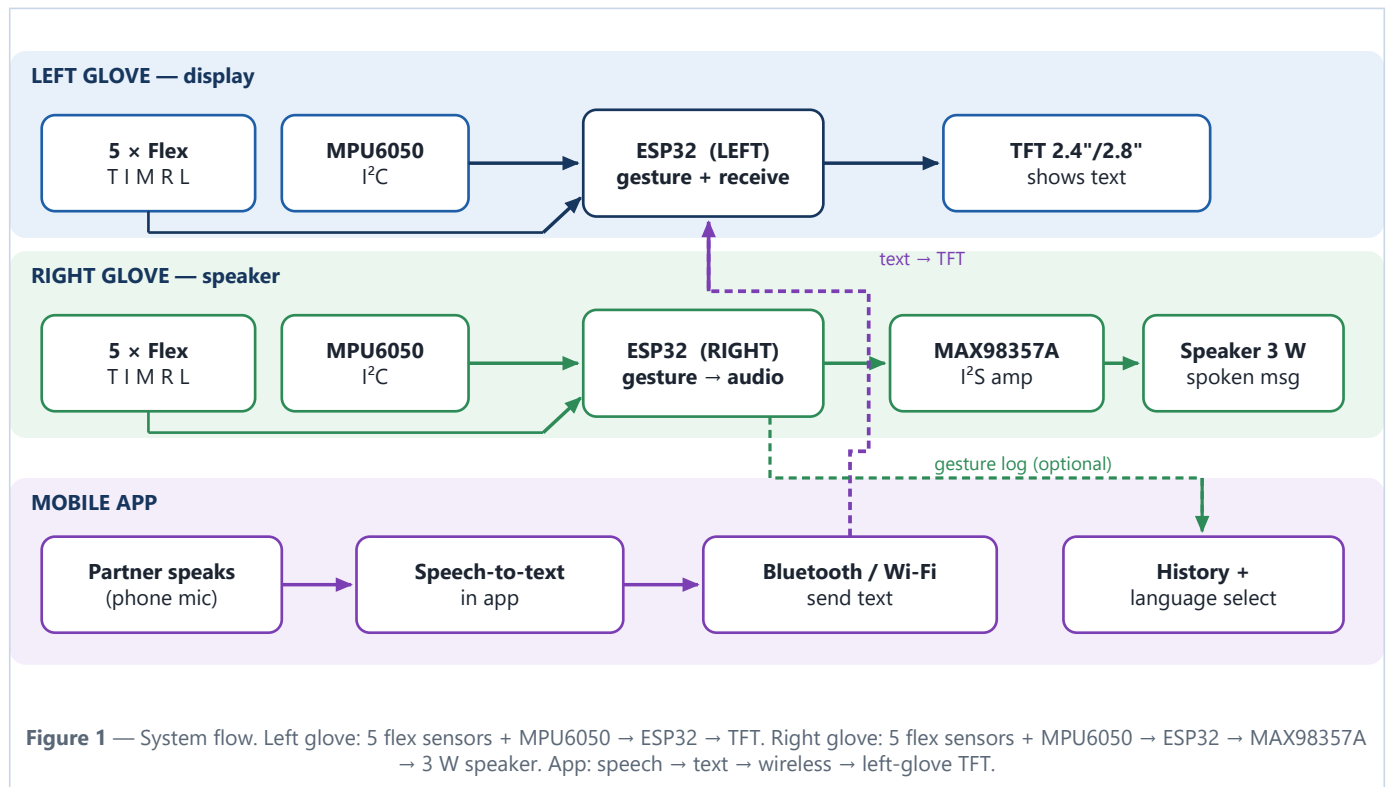
A practical, picture-by-picture guide for assembling, wiring, programming, calibrating and testing the two smart gloves: the **left display glove** and the **right speaker glove**.



The finished prototype: left display glove, right speaker glove, components, inside views and the mobile app.

Important: this guide keeps the exact component roles from your project. The ESP32 GPIO numbers shown in the diagrams are a **suggested, tested-in-common-use pin map** (flex sensors on ADC1 pins, MPU6050 on the default I²C pins, TFT on common TFT_eSPI pins, MAX98357A on free I²S pins). Pin labels differ between boards: check the labels printed on your modules and their datasheets before making permanent connections.

OVERVIEW — WHAT YOU ARE BUILDING



Left hand glove (display)



Right hand glove (speaker)

Bill of materials

Component	Qty	Used in	Role
ESP32 dev board	2	Both gloves	Reads sensors, recognises gestures, drives output
Flex sensor (2.2" / 4.5")	10	5 per glove	Measures bend of each finger
MPU6050 (IMU)	2	1 per glove	Hand tilt and rotation (I ² C)
TFT display 2.4" / 2.8"	1	Left glove	Shows received text / gestures
MAX98357A I ² S amplifier	1	Right glove	Drives the speaker from ESP32 I ² S
Speaker 3 W	1	Right glove	Speaks recognised gestures
3.7 V Li-ion battery	2	1 per glove	Power source
TP4056 charger module	2	1 per glove	Safe charging (use version with protection)
10 kΩ resistor	10	5 per glove	Fixed half of each flex voltage divider
Perfboard / PCB	2	1 per glove	Mount dividers and connectors
Switch	2	1 per glove	Main power on/off
Gloves, jumper wires	2 / as req.	—	Base and interconnects

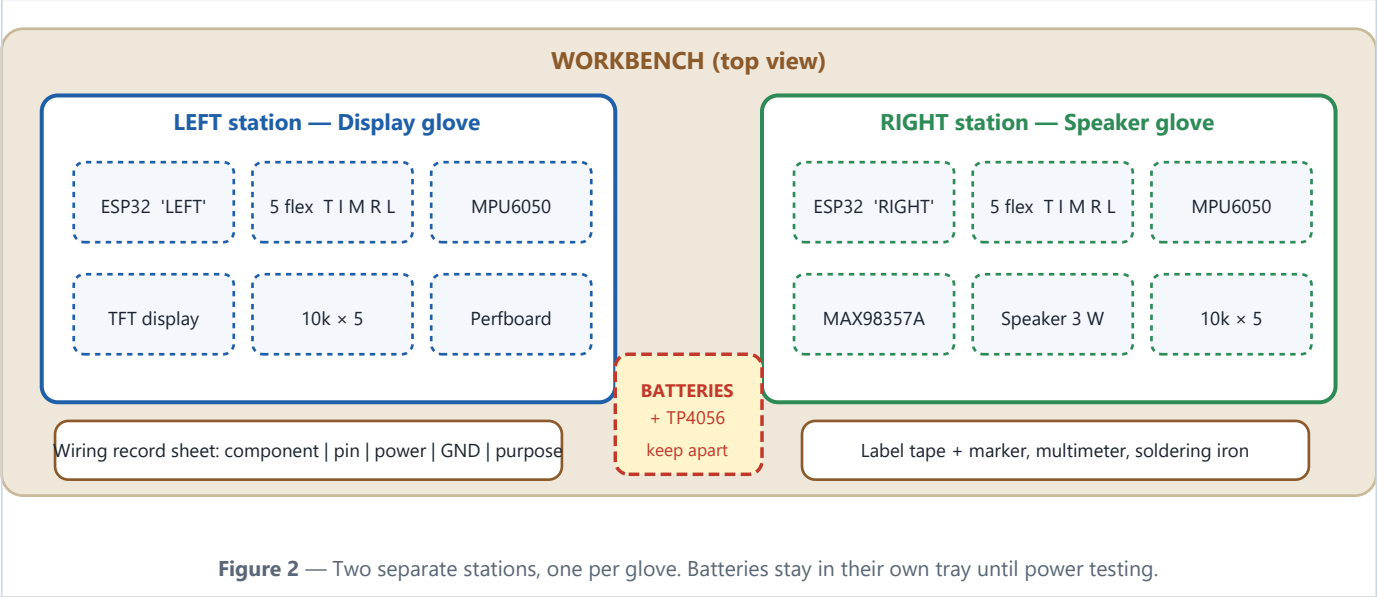


Main components (from the prototype design sheet)

Suggested ESP32 pin map (verify against your boards)

Signal	Left glove ESP32	Right glove ESP32	Why this pin
Flex T / I / M / R / L	GPIO36, 39, 34, 35, 32	GPIO36, 39, 34, 35, 32	ADC1 pins keep working while Wi-Fi/BT is on (ADC2 does not)
MPU6050 SDA / SCL	GPIO21 / GPIO22	GPIO21 / GPIO22	Default ESP32 I ² C pins (Wire library)
TFT CS / DC / RST	GPIO15 / GPIO2 / GPIO4	—	Common TFT_eSPI ESP32 setup
TFT MOSI / SCK / MISO	GPIO23 / 18 / 19	—	ESP32 VSPI hardware bus
I ² S BCLK / LRC / DIN	—	GPIO26 / 25 / 27	Free output pins, not used by other modules
Power	3V3 → sensors, TFT	5 V → MAX98357A, 3V3 → sensors	Common ground everywhere

1. WORKSPACE SETUP



- 1

Make two work areas

Put the left-glove components on one side and the right-glove components on the other. Keep batteries separate until power testing.
- 2

Label everything

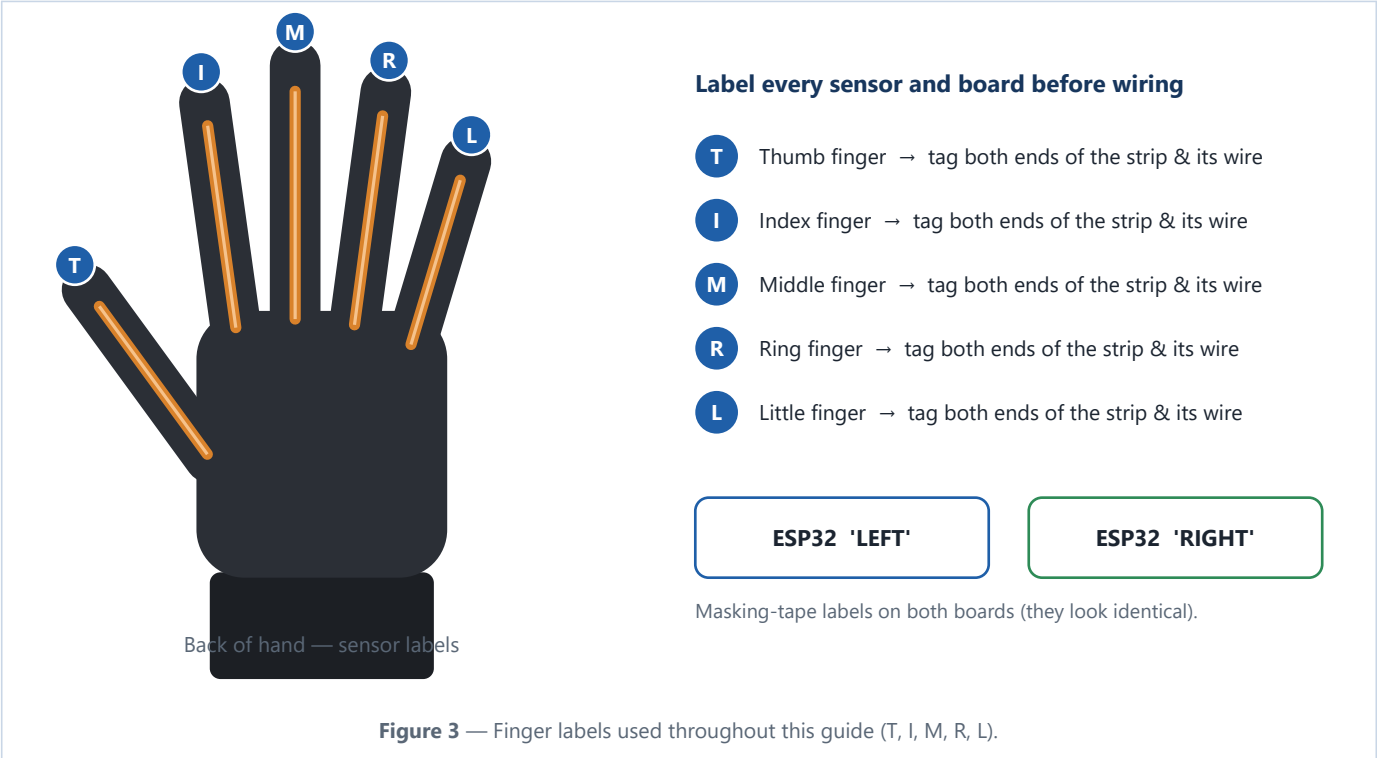
Label the five flex sensors T, I, M, R, L for thumb, index, middle, ring and little finger. Label the two ESP32 boards LEFT and RIGHT.
- 3

Prepare a wiring record

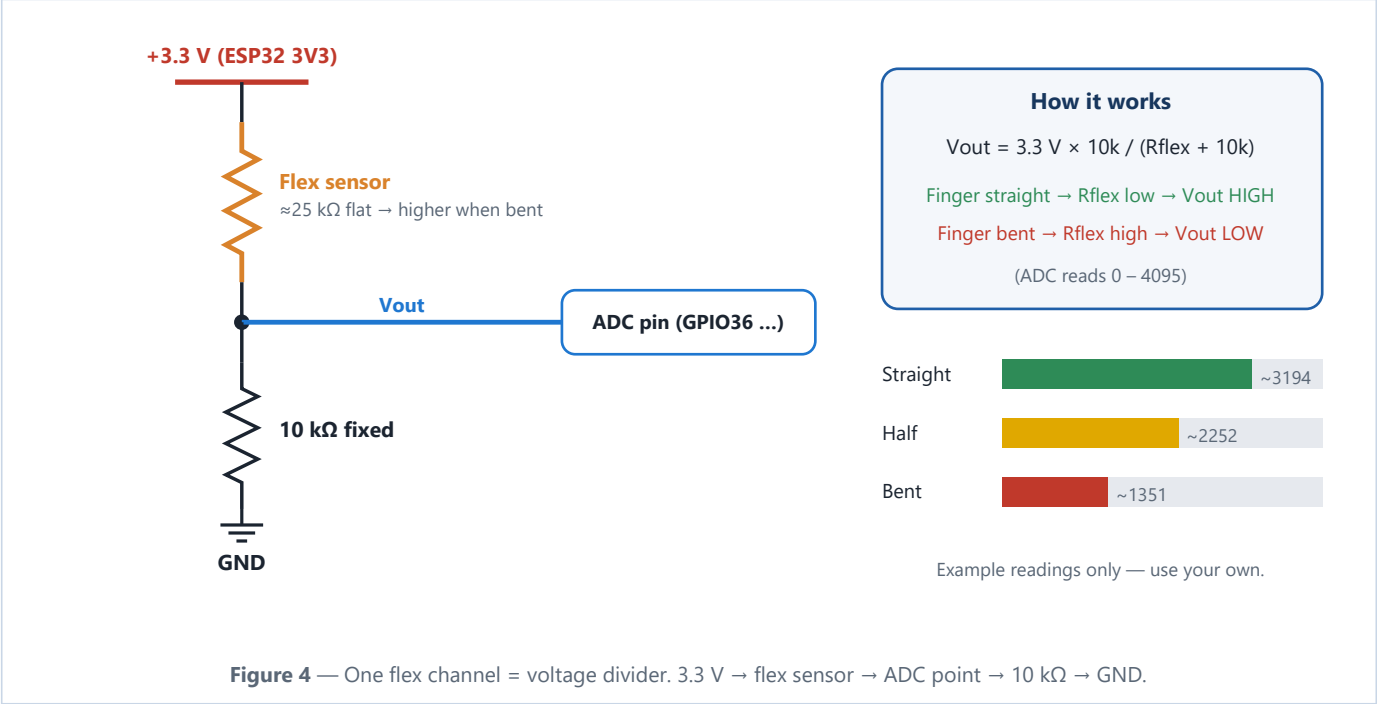
Draw a table with component, ESP32 pin, power, ground and purpose. Do not rely on memory when many wires are connected.
- 4

Keep power disconnected

During signal wiring, keep batteries and USB power disconnected unless you are intentionally testing the circuit.



2. FLEX SENSOR WIRING AND DIRECTION



- 5 Understand the flex sensor**
A flex sensor changes resistance when it bends. The ESP32 cannot directly measure resistance, so use a voltage divider.
- 6 Build one test channel first**
Connect 3.3 V to one end of the flex sensor. Connect its other end to the ADC measurement point. Connect a fixed resistor from the ADC point to GND.
- 7 Connect the ADC point**
Connect the voltage-divider midpoint to an ADC-capable ESP32 input. Keep the sensor signal wire short and secure.
- 8 Repeat for five fingers**
Build five separate voltage-divider channels. Never connect multiple sensor outputs together.

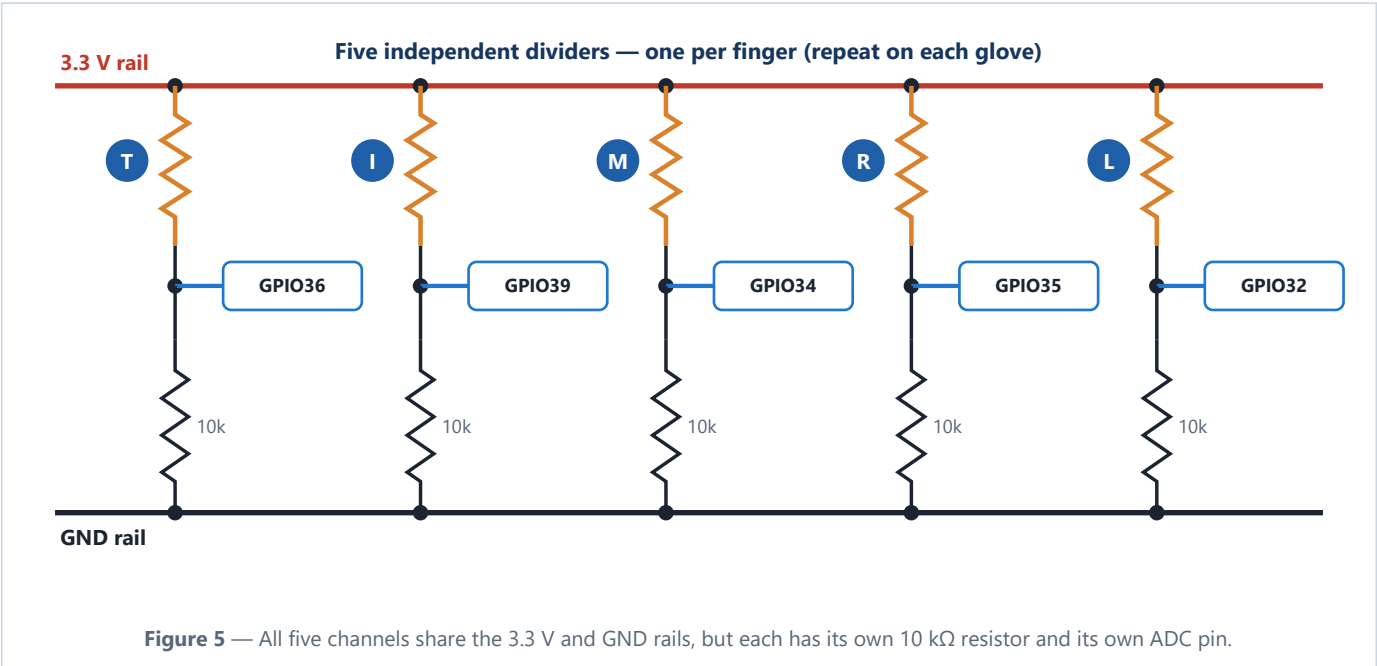




Figure 6 — Mounting a flex sensor: fix the tip, let the strip slide, protect the base pins.

- 9

Mount the sensors
Place each sensor along the finger so it bends naturally with the finger. Keep the sensor's sensitive section away from sharp folds.
- 10

Check orientation
Before fixing the sensor, bend the finger slowly. Make sure the sensor bends in the intended direction and the cable does not pull on the sensor.
- 11

Test one sensor
Power the ESP32 and print the ADC reading. Record the value with the finger straight, half bent and fully bent.
- 12

Calibrate all five
Repeat the measurement for thumb, index, middle, ring and little finger. Take several readings for each position.
- 13

Create thresholds
Use the observed ranges to define straight, bent and intermediate states. Use ranges rather than a single exact number.

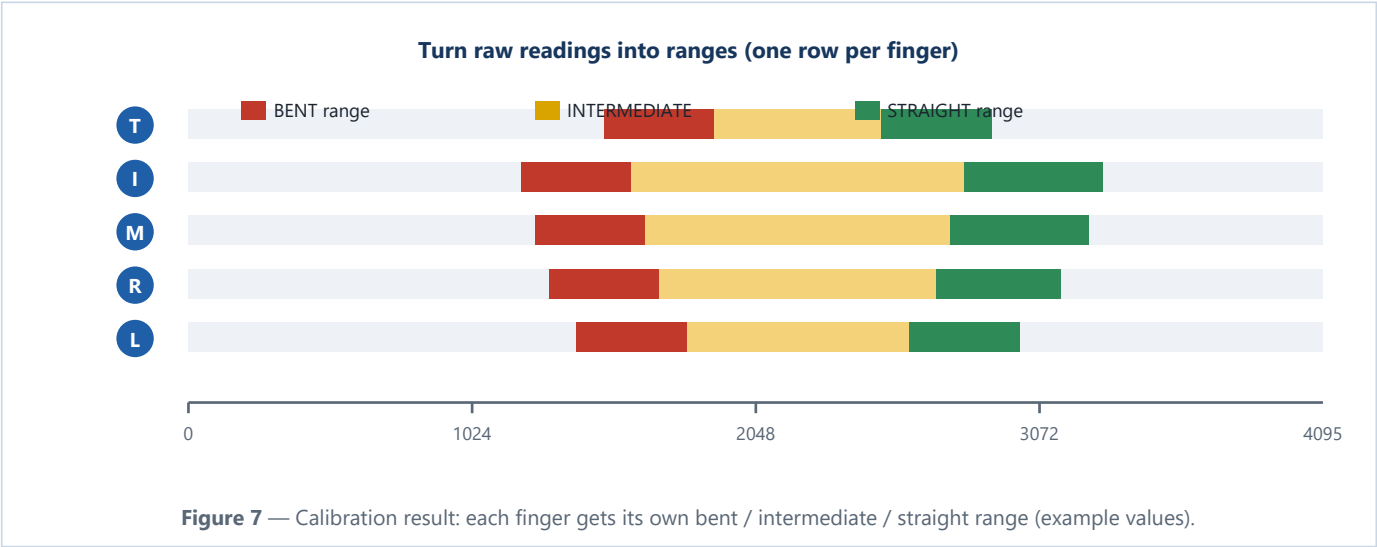


Figure 7 — Calibration result: each finger gets its own bent / intermediate / straight range (example values).

3. MPU6050 WIRING AND DIRECTION

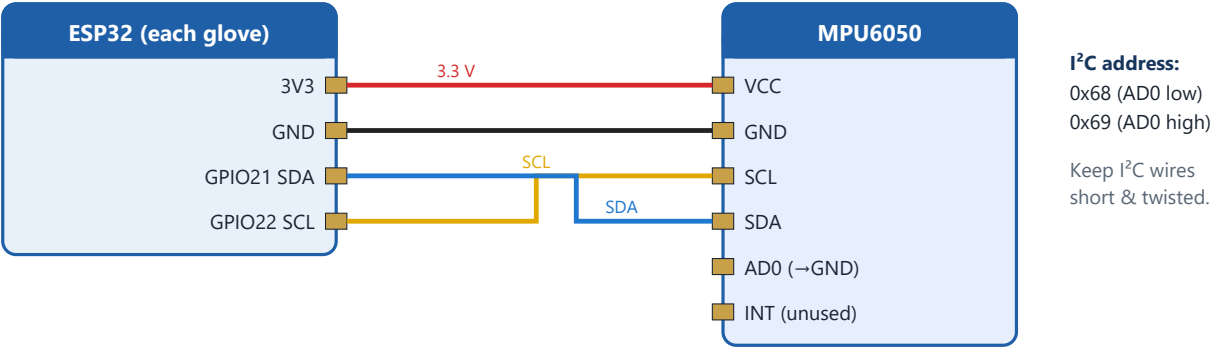


Figure 8 — MPU6050 to ESP32 over I²C — same wiring on both gloves.

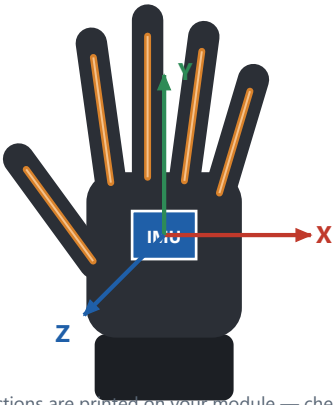
- 14

Identify the MPU6050 pins
Find VCC, GND, SDA and SCL on your actual module. Follow the module's voltage requirements.
- 15

Connect power
Connect the module's required supply and GND. For a typical ESP32 setup, the sensor logic is commonly operated at 3.3 V, but verify your module.
- 16

Connect I²C
Connect SDA to the ESP32 SDA pin and SCL to the ESP32 SCL pin selected in your firmware.
- 17

Run an I²C scan
Upload an I²C scanner and verify that the MPU6050 appears at its expected address. If it does not, check power, ground and SDA/SCL wiring.



Axis directions are printed on your module — check them.

Figure 9 — Place the MPU6050 flat on the back of the hand and check each axis responds.

Movement test	What you should see
---------------	---------------------

- | | |
|-----------------------|---------------------------|
| Hold still, palm down | AZ ≈ +1 g, GX/GY/GZ ≈ 0 |
| Tilt forward / back | AY changes sign (pitch) |
| Tilt left / right | AX changes sign (roll) |
| Twist wrist | GZ spikes while rotating |
| Still for 5 s | record offsets → subtract |

- 18

Test acceleration
Print AX, AY and AZ. Keep the glove still, then tilt it forward, backward and sideways.
- 19

Test rotation
Print GX, GY and GZ. Rotate the glove around different axes and confirm that the readings respond.
- 20

Calibrate
Keep the glove motionless and calculate or record the offsets required by your software.

21

Repeat for the second glove

Perform the same wiring, address check and calibration on the right glove.

4. LEFT DISPLAY GLOVE



Outside: TFT on the back of the hand



Inside view: flex wires, TFT, ESP32, battery

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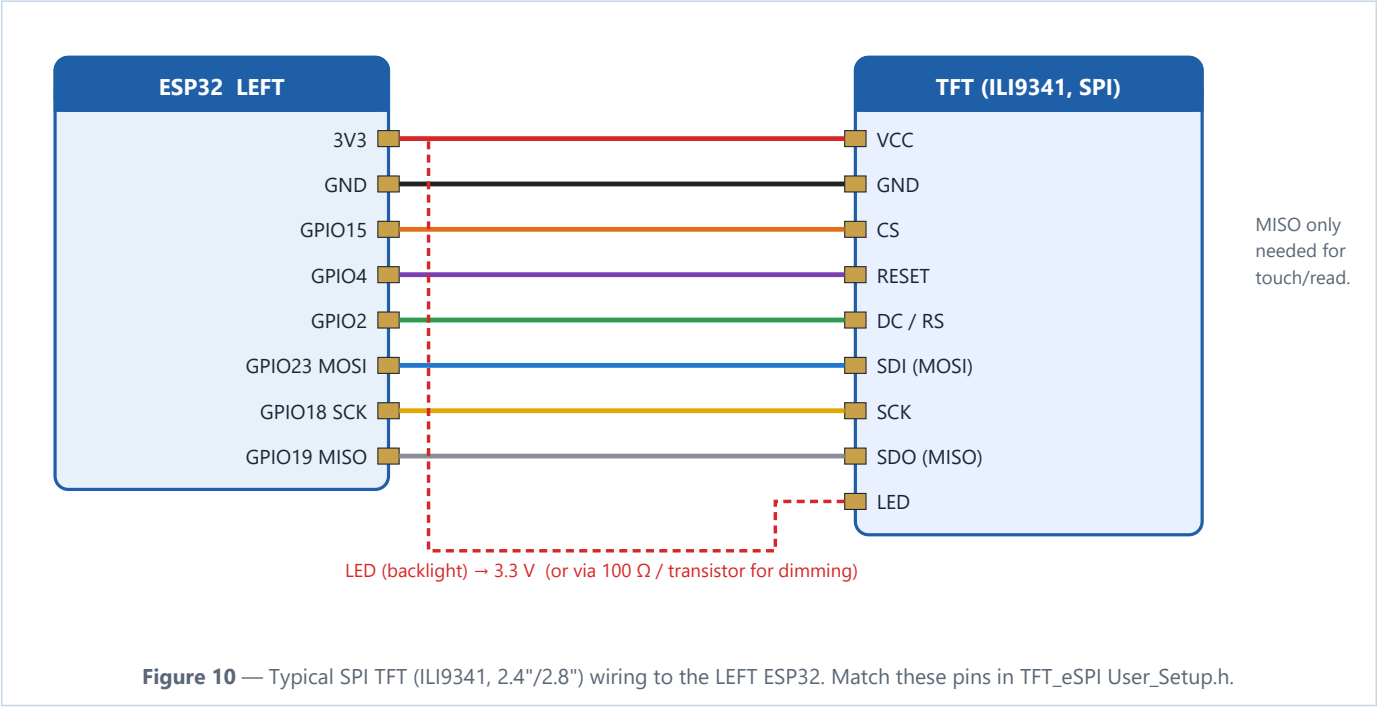
Plan physical placement

Place the ESP32 where it will not interfere with the wrist. Place the TFT on the back of the hand where it is visible to the wearer.

23

Wire the TFT

Identify the TFT interface and connect power, ground and its communication/control pins according to the exact module pinout.



24

Test the display alone

Upload a simple program that displays HASTH VANI, READY and a test message.

- 25

Add flex and MPU6050 inputs
Run the sensor-reading program and display the live sensor values or a short diagnostic screen.
- 26

Create a gesture record
Combine five flex readings and the relevant MPU6050 orientation values into one data structure.
- 27

Recognize one gesture
Start with HELLO. Record several examples and identify the sensor ranges that distinguish it.
- 28

Add the message
When HELLO is recognized, display HELLO on the TFT.
- 29

Add more gestures
Add WATER, HELP, THANK YOU, YES, NO and other predefined messages chosen by your team.
- 30

Add an unknown state
If no gesture matches reliably, display UNKNOWN or READY rather than producing a random message.

Gesture	T	I	M	R	L	Orientation	Output
HELLO	S	S	S	S	S	palm forward, upright	"Hello"
WATER	B	S	S	S	B	tilted, index to mouth	"I need water"
HELP	S	B	B	B	B	thumb up, wrist raised	"Help"
THANK YOU	S	S	S	S	S	move away from chin	"Thank you"
YES	B	B	B	B	B	fist, nod (pitch)	"Yes"
NO	½	S	S	B	B	wave side to side (GZ)	"No"

S = straight

½ = half bent

B = bent

EXAMPLE template — fill it from YOUR recorded samples.

Figure 11 — A gesture record = 5 finger states + orientation → message. Build yours from real samples.

5. RIGHT SPEAKER GLOVE



Outside: speaker, amplifier, ESP32 on cuff



Inside view: MPU6050, ESP32, amplifier, speaker, battery

31	Mount the second ESP32 Place it securely and leave enough room for the USB connector and wiring.
32	Mount the speaker Position the speaker so its sound outlet is not covered by the glove or mounting material.
33	Connect MAX98357A Identify VIN/VCC, GND, BCLK, LRC/WS and DIN on the exact amplifier board. Connect the ESP32 I ² S signals according to your firmware and module.
34	Connect speaker Connect the speaker to the amplifier's speaker output using the module's marked terminals. Check polarity and avoid shorting the outputs.

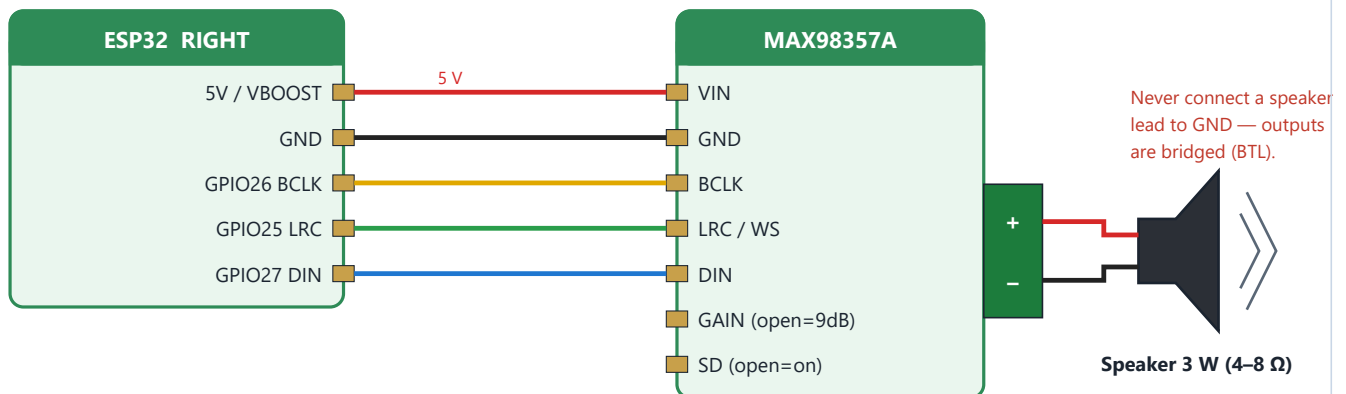


Figure 12 — MAX98357A I²S amplifier and 3 W speaker wiring to the RIGHT ESP32.

35	Test audio Before adding gesture recognition, play a known audio test. Verify volume, clarity and temperature.
36	Add the second glove sensors Connect its five flex sensors and MPU6050 using the same approach as the display glove (Figures 5 and 9).
37	Recognize one gesture Test HELLO first. Confirm that the second ESP32 identifies the intended gesture.
38	Trigger audio Map the recognized gesture to the corresponding audio message and play it through the MAX98357A and speaker.

6. GESTURE RECOGNITION DIRECTIONS

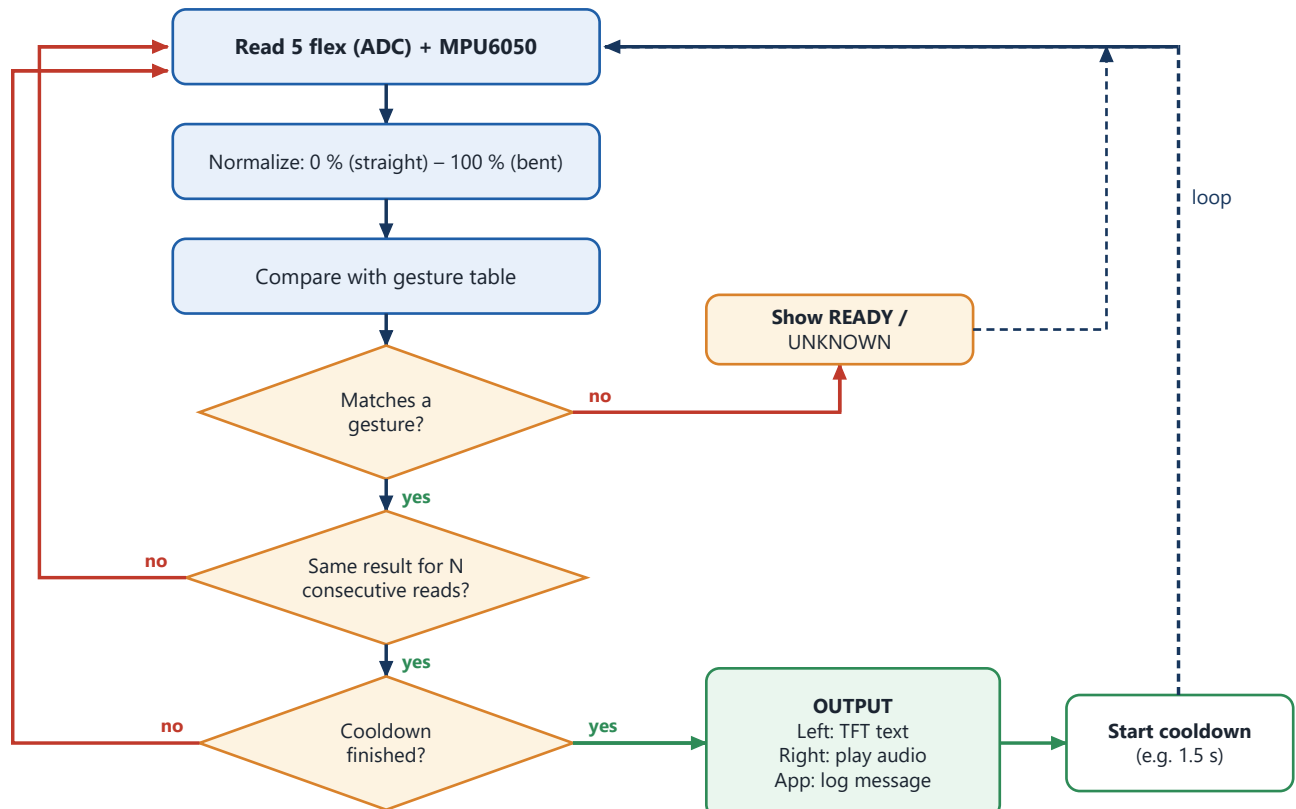


Figure 13 — Recognition loop used on both gloves: stable readings + cooldown prevent random and duplicate messages.

39 Collect real samples

Wear the glove and perform each intended gesture repeatedly. Save the sensor values for successful and unsuccessful attempts.

40 Normalize sensor behavior

If needed, map each flex reading to a normalized range using its calibrated straight and bent limits.

41 Use multiple samples

Do not accept a gesture from one noisy reading. Check several consecutive readings and accept the gesture only when they remain consistent.

42 Combine hand shape and orientation

Use both flex-sensor states and MPU6050 orientation where necessary so similar finger patterns can be separated.

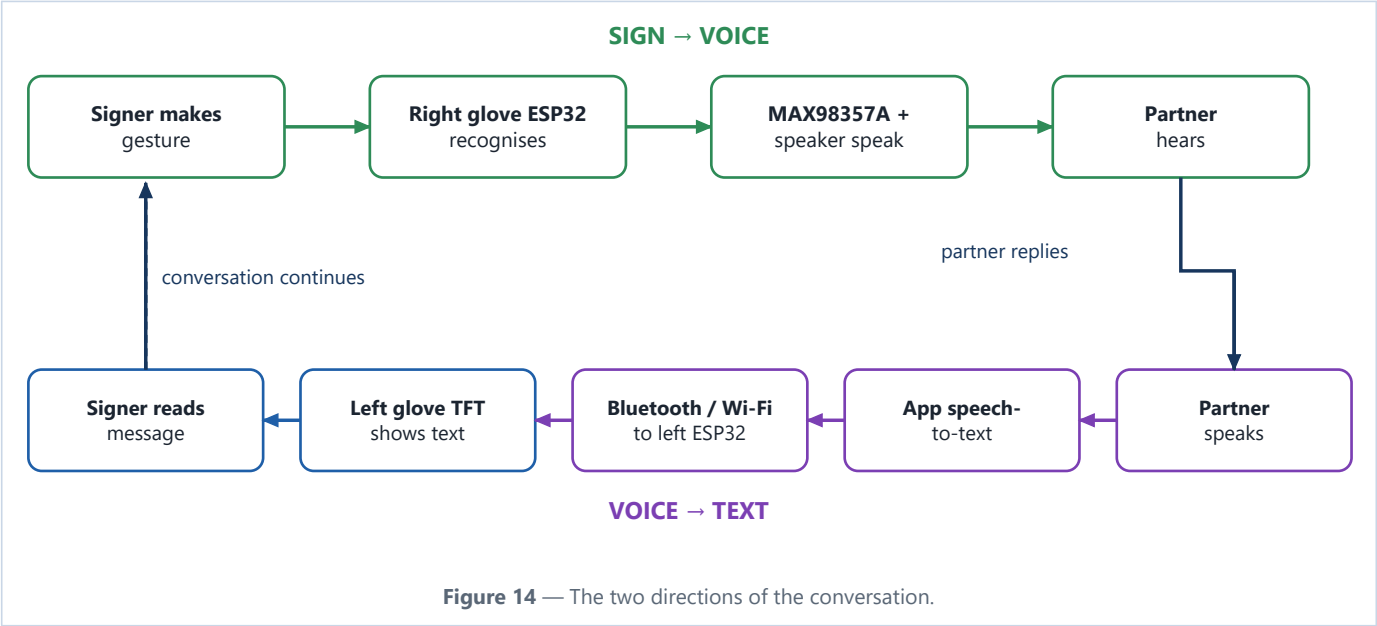
43 Add a cooldown

After a gesture is accepted, briefly prevent immediate repeated triggering. This reduces duplicate messages.

44 Test transitions

Move from one gesture to another and check whether the system incorrectly recognizes intermediate positions.

7. MOBILE APP AND TWO-WAY COMMUNICATION



App screens

Left screen — glove to app: connection status, the detected gesture (e.g. I NEED WATER with a confidence value), a Speak button, language selection and a translation history.

Right screen — app to glove: tap the microphone, the partner speaks, the recognised text appears ("Do you need water?"), then **Send to Left Glove** pushes it to the TFT.

- 45

Connect the app
Use the wireless method already selected for your project, such as Bluetooth or Wi-Fi.
- 46

Show connection status
Display connected/disconnected state so the user knows whether communication is available.
- 47

Send recognized messages
Send the recognized gesture/message to the app for history or display.
- 48

Implement speech-to-text
Capture speech from the communication partner's phone microphone and convert it into text using the app's speech-recognition function.
- 49

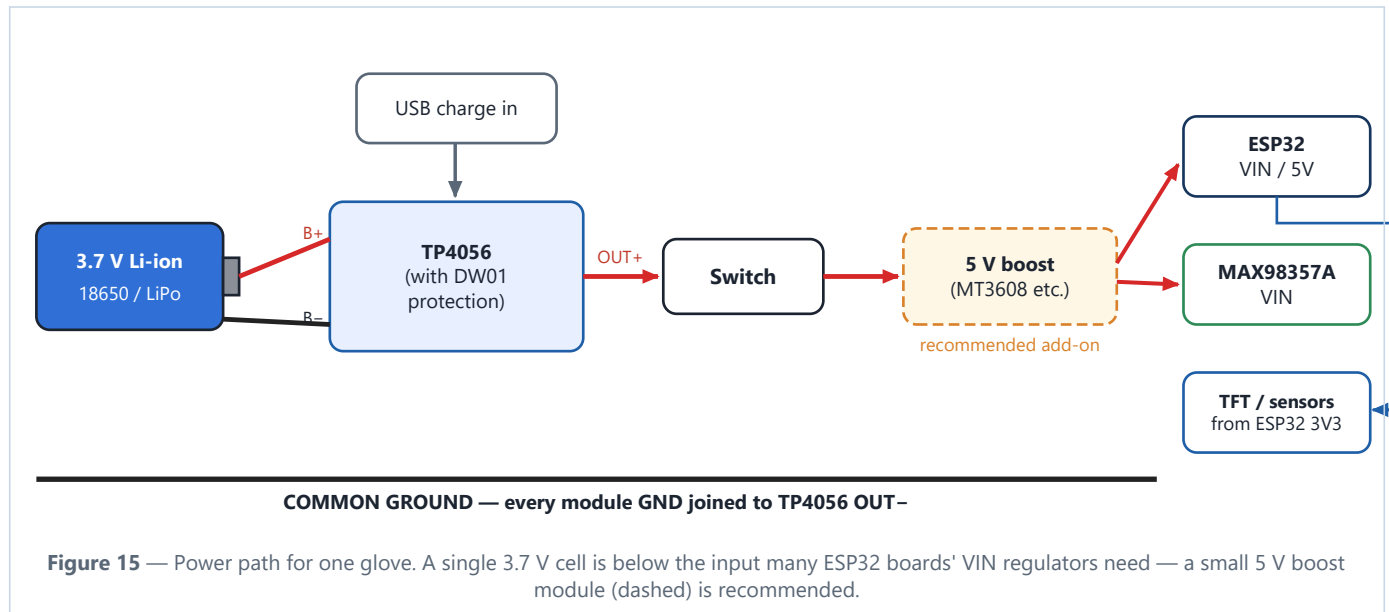
Send text to the left glove
Transmit the resulting text to the display glove.
- 50

Display received text
Show the received message clearly on the TFT.

51 Test the complete reverse direction

Speak a short sentence, verify the app's transcription, verify wireless transmission and confirm that the TFT displays the result.

8. POWER AND SAFE PHYSICAL ASSEMBLY

**52 Test power modules separately**

Check the TP4056 charging module and battery arrangement before connecting the complete glove.

53 Insulate exposed joints

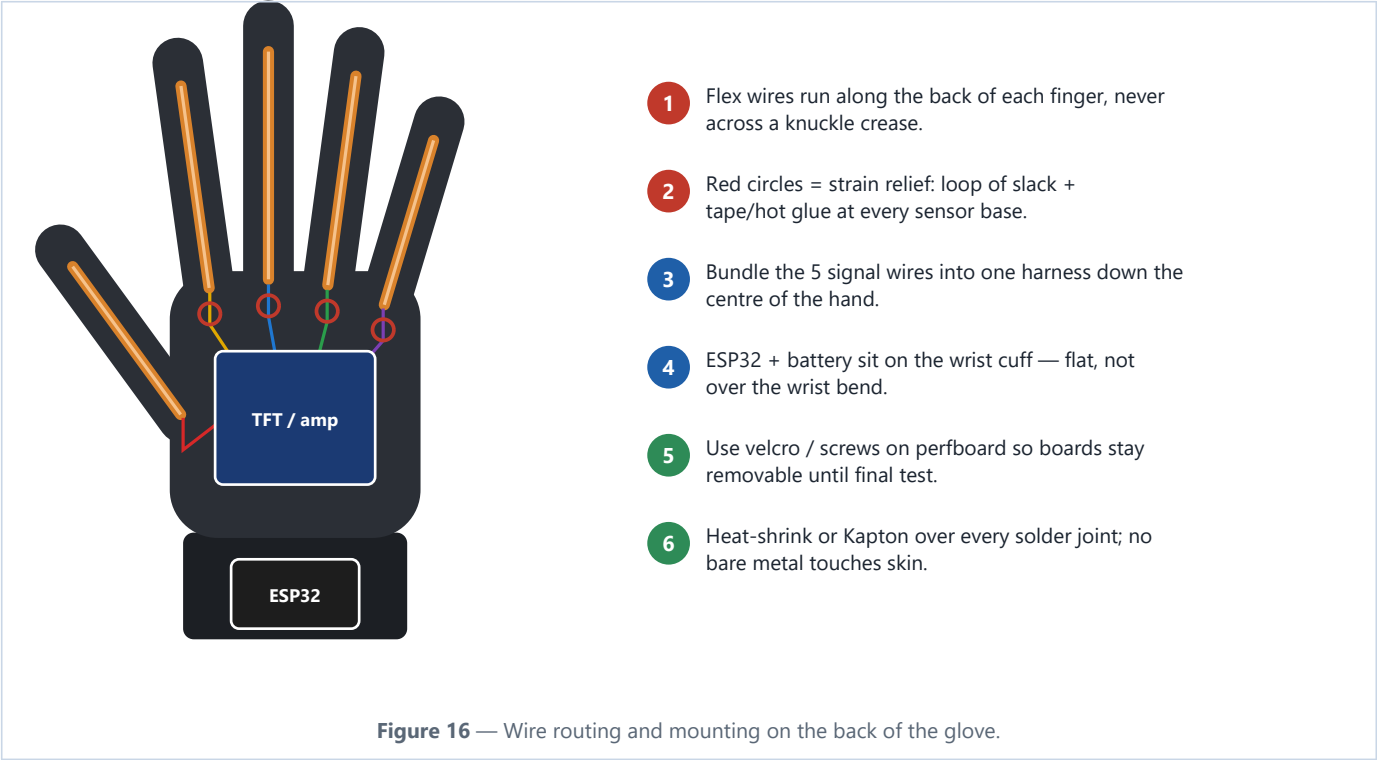
Cover solder joints and exposed conductors with suitable insulation. Prevent accidental shorts.

54 Add strain relief

Secure wires near flex sensors, the wrist and board connectors so movement does not pull solder joints.

55 Route wires

Keep wires along the glove surface and away from moving finger joints as much as possible.



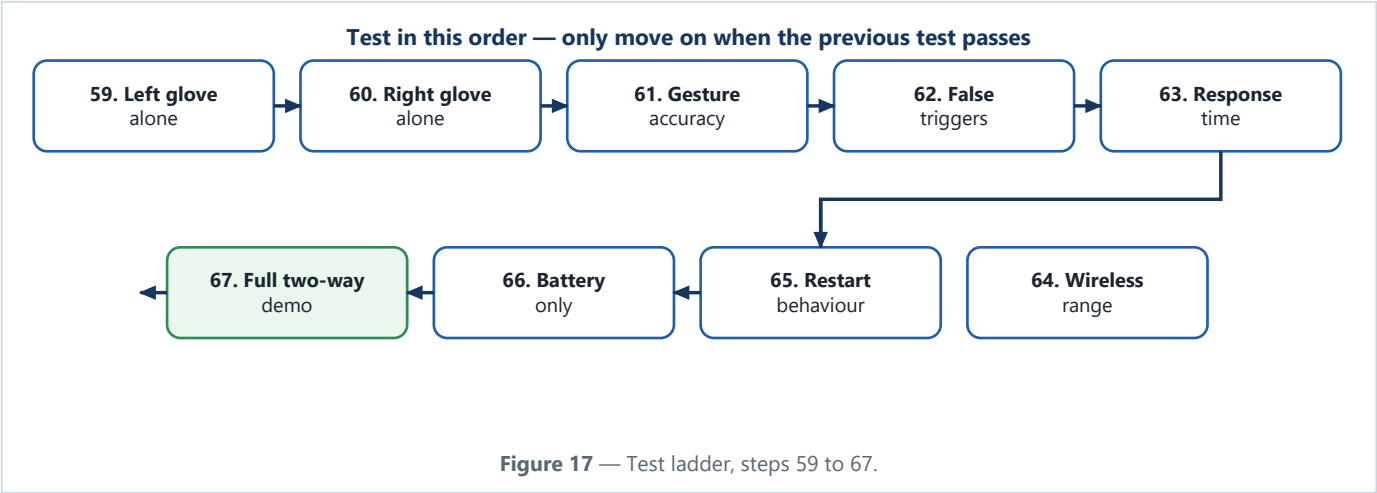
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Secure the boards
Use a secure but serviceable mounting method. Do not permanently seal the electronics until final testing is complete.
- 57

Check comfort
Wear the glove and slowly move every finger. Confirm that no board, wire or battery causes uncomfortable pressure.
- 58

Check heat
During a full-power test, touch-check only where safe and monitor components for abnormal heating. Disconnect power if anything becomes unexpectedly hot.

9. COMPLETE TESTING



- 59

Test left glove alone
Run all five flex sensors, MPU6050 and TFT together.
- 60

Test right glove alone
Run all five flex sensors, MPU6050, amplifier and speaker together.

61	Test gesture accuracy Perform every supported gesture multiple times and record correct and incorrect recognitions.
62	Test false triggers Move naturally without intentionally making a gesture. Check whether the system produces unwanted messages.
63	Test response time Measure the approximate time from completing a gesture to seeing or hearing the output.
64	Test wireless range Walk the distance required for your demonstration and verify that communication remains stable.
65	Test restart behavior Turn the system off and on again. Confirm that both gloves return to a usable state.
66	Test battery operation Run the prototype from its intended battery arrangement and observe operating stability.
67	Test the two-way demo Gesture → recognition → voice/text, then spoken response → speech-to-text → wireless message → TFT.

10. FINAL ASSEMBLY

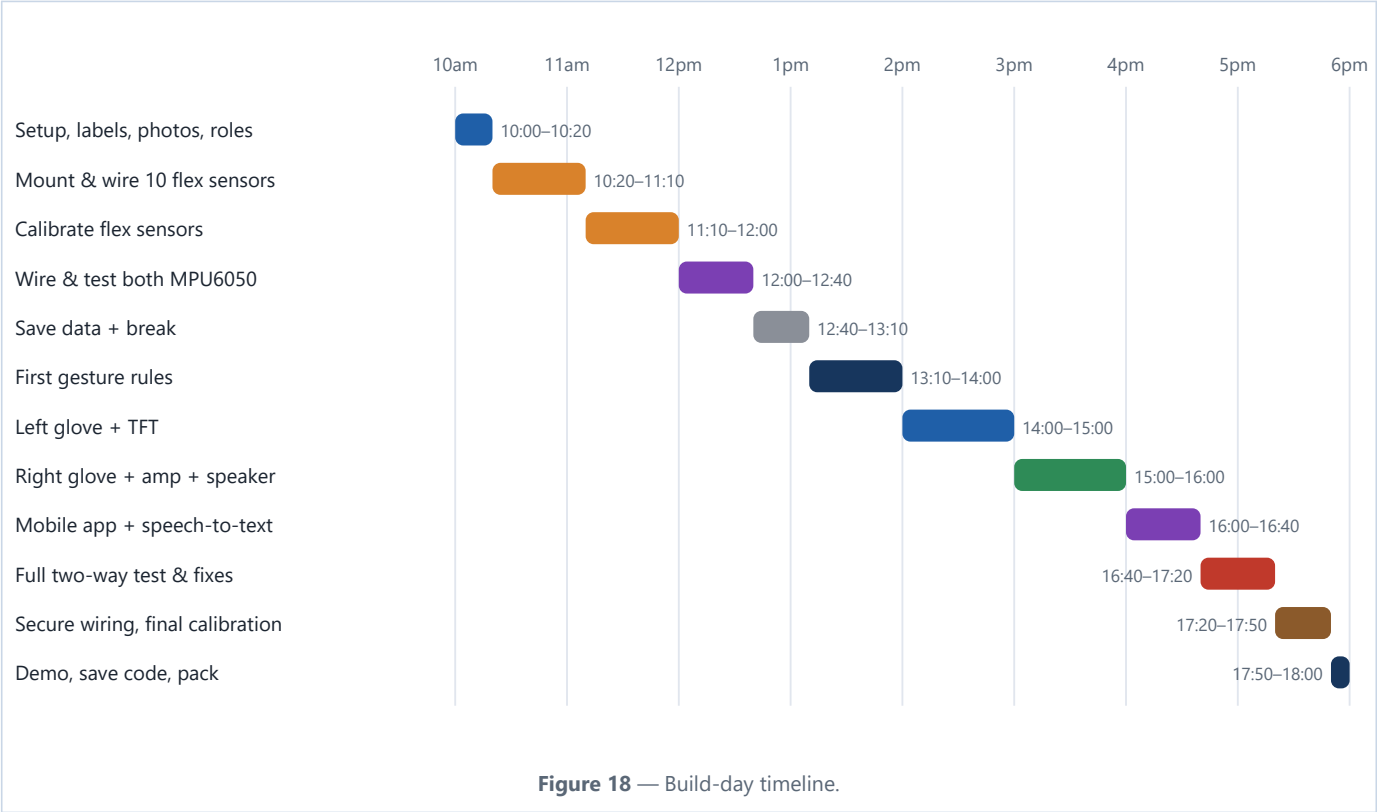


Target look: both gloves worn

Close-ups: flex sensor, MPU6050, TFT, speaker

68	Freeze the wiring After all tests pass, photograph the wiring and document every connection before final dressing.
69	Secure components Fix the ESP32, sensors, display, amplifier and speaker in their final positions.
70	Label important wires Label power, ground, I ² C and audio lines where practical to simplify future repairs.
71	Protect connectors Secure loose connectors so normal glove movement does not disconnect them.
72	Final calibration Perform one final calibration while wearing the finished glove, because mounting position can change sensor readings.
73	Save firmware Keep a copy of the final ESP32 code and the last known-good version.
74	Record limitations Document the supported gestures, approximate recognition behavior, battery limitations and known failure cases.

11. 10 AM–6 PM BUILD-DAY DIRECTIONS



10:00–10:20	Set up two workstations, label components, photograph starting condition and assign team roles.
10:20–11:10	Mount and wire the ten flex sensors. Test each sensor before permanent fixing.
11:10–12:00	Calibrate all flex sensors and record threshold values.
12:00–12:40	Wire and test both MPU6050 modules.
12:40–1:10	Save calibration data and take a short break.
1:10–2:00	Implement and test the first gesture-recognition rules.
2:00–3:00	Integrate the left glove with TFT and test displayed messages.
3:00–4:00	Integrate the right glove with MAX98357A and speaker.
4:00–4:40	Connect the mobile application and implement/test speech-to-text communication.
4:40–5:20	Run the complete two-way system and fix the most important failures.
5:20–5:50	Secure wiring, mount components and perform final calibration.
5:50–6:00	Run the demonstration once, save code/data, photograph the final prototype and pack all parts.

TEAM ROLE PLAN

Member	Primary responsibility
1	Left-glove flex sensors, calibration and physical mounting
2	Right-glove flex sensors, calibration and physical mounting
3	ESP32 firmware and gesture-recognition logic
4	TFT/display glove and wireless message display
5	MAX98357A, speaker and right-glove audio system
6	Mobile app, speech-to-text, testing and documentation

FINAL CHECKLIST

- | | |
|--|---|
| ☐ All 10 flex sensors tested and calibrated. | ☐ Text reaches the display glove. |
| ☐ Both MPU6050 modules detected and calibrated. | ☐ Two-way communication demonstrated. |
| ☐ Left ESP32 + TFT working. | ☐ Battery and charging system checked safely. |
| ☐ Right ESP32 + MAX98357A + speaker working. | ☐ Wires secured and exposed connections insulated. |
| ☐ Gesture recognition tested for every supported gesture. | ☐ Final firmware backed up. |
| ☐ Unknown/no-gesture state works. | ☐ Supported gestures and limitations documented. |
| ☐ Mobile speech-to-text works. | ☐ |

TROUBLESHOOTING QUICK GUIDE

Problem	First checks	See
Flex values do not change	Check voltage divider, ADC pin, ground, sensor orientation and wire continuity.	Fig. 4–6
MPU6050 not detected	Check VCC/GND, SDA/SCL, I ² C address and pull-ups/module configuration.	Fig. 8
TFT blank	Check power, ground, interface type, CS/DC/RST pins and display initialization.	Fig. 11
Speaker silent	Check I ² S pins, MAX98357A power, speaker connection and audio sample configuration.	Fig. 13
Random gestures	Recalibrate flex sensors, widen/narrow thresholds appropriately, use multiple samples and add a cooldown.	Fig. 7, 14
Wireless connection drops	Check power stability, distance, reconnection logic and whether both devices use the same communication settings.	Fig. 15, 16
ESP32 resets when speaker plays	Battery voltage sagging: add the 5 V boost, a 470 μ F capacitor near the amp, and shorter/thicker power wires.	Fig. 16